Integer Factorization Complexity

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Subject: Proposed Hybrid Framework for Sub-Exponential Integer Factorization via Algebraic Number Theory and Quantum-Inspired Classical Optimization

1. Core Hypothesis

The hardness of integer factorization ($N = p \times q$) does not stem solely from the combinatorial explosion of trial division, but from the inability to efficiently map the multiplicative group structure $\mathbb{Z}_N^*$ onto a lattice with low enough density to permit polynomial-time sieving using classical resources.

Hypothesis: By constructing a generalized algebraic number field extension $K(\sqrt{-D})$ where D is derived from the specific bit-pattern entropy of N, we can construct a "smoothness manifold" that allows for the reduction of the General Number Field Sieve (GNFS) complexity exponent from $O(e^{(64/9)^{1/3} (\log N)^{1/3} (\log \log N)^{2/3}})$ to $O(e^{c (\log N)^{1/3} (\log \log N)^{1/3}})$, effectively bridging the gap between current classical limits and quantum polynomial time without requiring fault-tolerant quantum hardware. This relies on the conjecture that certain high-dimensional algebraic lattices associated with semi-primes exhibit hidden symmetries exploitable via classical randomized walks.

2. Mathematical Framework

Let N be an odd composite integer. We propose a modified Number Field Sieve (M-NFS) operating within a dual-field architecture:

* **Field Construction:** Instead of standard irreducible polynomials $f(x), g(x)$ used in GNFS, we define a dynamic ring extension $\mathcal{O}_{K_N}$ over a quadratic imaginary field $K = \mathbb{Q}(\sqrt{-\Delta_N})$, where Δ_N is selected based on the Jacobi symbol properties of N.

$$ \Delta_N = \text{argmin}_{d} \left(\lambda_d(N) \right) $$

where $\lambda_d(N)$ represents the estimated smoothness probability of elements generated by ideals in the ring $\mathbb{Z}[\sqrt{-d}]$ modulo N.

* **Lattice Mapping:** We map the logarithms of prime ideals in $\mathcal{O}_{K_N}$ to a high-dimensional Euclidean space $\mathbb{R}^k$ via the Minkowski embedding. The core innovation is the definition of a *Smoothness Potential Function* Φ:

$$ \Phi(v) = \sum_{i=1}^k w_i \cdot \log |v_i| - \epsilon \cdot \text{Entropy}(v) $$

Where w_i are weights derived from the distribution of small primes in the residue system of N.

* **Optimization Objective:** The problem reduces to finding a non-trivial kernel element in the ideal class group $Cl(K_N)$ such that the product of principal ideals yields a square relation:

$$ \prod \mathfrak{p}_j^{e_j} = \alpha^2 \quad \text{and} \quad \prod \bar{\mathfrak{p}}_j^{e_j} = \beta^2 $$

leading to $x^2 \equiv y^2 \pmod N$ with $x \not\equiv \pm y \pmod N$.

3. Proof Sketch / Key Steps

Step 1: Entropy-Guided Polynomial Selection

Standard GNFS uses heuristic polynomial selection. We replace this with a deterministic search for polynomials minimizing the root-finding variance relative to the bit-depth of N.

* *Sketch:* Define a cost function $C(f) = \int |f(x)| dx$ weighted by the probability of $f(x)$ being B-smooth. Prove that for large N, there exists a family of polynomials F_N where the average value magnitude scales as $(\log N)^{1-\delta}$ rather than the standard exponential growth, reducing the sieve range required.

Step 2: Accelerated Relation Collection via Random Walks

Utilize a Metropolis-Hastings algorithm adapted for the ideal class group.

* *Sketch:* Construct a Markov Chain Monte Carlo (MCMC) process on the graph of ideals. The transition probabilities are biased towards ideals whose norms have high densities of small prime factors (smooth numbers).

* *Key Lemma:* If the mixing time of this chain is bounded by $poly(\log N)$, then the collection of relations (pairs (a, b) such that $a^2 \equiv b^2 \pmod N$) can be achieved in sub-exponential time. This depends on the spectral gap of the adjacency matrix of the ideal graph being inversely proportional to $\log N$.
Step 3: Linear Algebra Reduction
Once sufficient relations are gathered, solve the linear dependency problem over $\mathbb{F}_2$.
* *Sketch:* Use sparse matrix Gaussian elimination (or Wiedemann's algorithm) optimized for the specific sparsity pattern induced by our entropy-weighted sieve. The dimension of the matrix will be reduced due to the higher yield of smooth relations per sieve step.
Step 4: Square Root Extraction and GCD
Compute $x = \sqrt{\prod a}$ and $y = \sqrt{\prod b}$ modulo N.
* *Conclusion:* Calculate $g = \gcd(x-y, N)$. With high probability, $1 < g < N$, yielding a non-trivial factor.
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